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# Beyond Language Modeling: An Exploration of Multimodal Pretraining

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## Abstract

The visual world offers a critical axis for advancing foundation models beyond language. Despite growing interest in this direction, the design space for native multimodal models remains opaque. We provide empirical clarity through controlled, from-scratch pretraining experiments, isolating the factors that govern multimodal pretraining without interference from language pretraining. We adopt the Transfusion framework, using next-token prediction for language and diffusion for vision, to train on diverse data including text, video, image-text pairs, and even action-conditioned video. Our experiments yield four key insights: (i) Representation Autoencoder (RAE) provides an optimal unified visual representation by excelling at both visual understanding and generation; (ii) visual and language data are complementary and yield synergy for downstream capabilities; (iii) unified multimodal pretraining leads naturally to world modeling, with capabilities emerging from general training; and (iv) Mixture-of-Experts (MoE) enables efficient and effective multimodal scaling while naturally inducing modality specialization. Through IsoFLOP analysis, we compute scaling laws for both modalities and uncover a scaling asymmetry: vision is significantly more data-hungry than language. We demonstrate that the MoE architecture harmonizes this scaling asymmetry by providing the high model capacity required by language while accommodating the data-intensive nature of vision, paving the way for truly unified multimodal models.

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## 1. Introduction

The foundation model era has been defined largely by the success of language pretraining (OpenAI, 2022; Google, 2023). By scaling autoregressive models on trillions of text tokens, we have created systems with remarkable reasoning capabilities. Yet, fundamentally, text is a human abstraction—a lossy compression of reality. To borrow the allegory of Plato’s Cave (Plato, 375 BCE), language models have mastered the description of shadows on the wall without ever seeing the objects casting them. They capture symbols well but miss the underlying physics and causality.

Beyond this philosophical limitation lies a hard, pragmatic ceiling: high-quality text data is finite and approaching exhaustion (Sutskever, 2025). In contrast, the visual world possesses an endless stream of signal “outside the cave”, capturing the raw dynamics of reality that language misses. As a result, the path forward requires moving beyond the shadows to model the source directly. We turn to unified multimodal pretraining, treating vision not as an auxiliary input but as a first-class signal alongside language.

The scientific landscape of unified multimodal pretraining remains largely opaque. While recent efforts (Wang et al., 2024; Zhou et al., 2025; Cui et al., 2025) have begun to move beyond language-only pretraining, the design space is rife with confounding variables. Most current methods (Liao et al., 2025; Deng et al., 2025) initialize from pretrained language models (Grattafiori et al., 2024; Yang et al., 2025), making it difficult to disentangle what is learned from unified training versus inherited from language pretraining. Consequently, the fundamental scaling dynamics between vision and language remain poorly understood.

In this work, we seek to provide empirical clarity to this landscape (Figure 1). We focus on pretraining, as a model’s core capabilities are largely acquired during this phase (Zhou et al., 2024). We train a single model from scratch using the Transfusion framework, with next-token prediction for language and diffusion for vision, on diverse data including text, video, image-text pairs, and action-conditioned video.

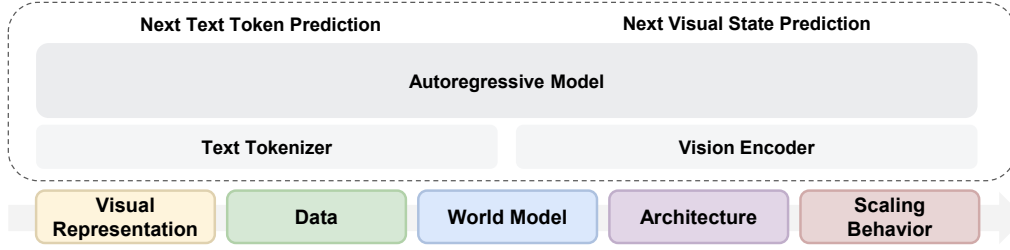


Figure 1. **Overview of study.** **Top:** High-level model architecture. We train an autoregressive model with next-token prediction for text and next-visual state prediction with flow matching. **Bottom:** We study representations, data, world modeling, architecture, and scaling.

We conduct controlled experiments to isolate key variables and evaluate across a broad spectrum of tasks.

Specifically, we investigate five axes: **(1) Visual Representation:** RAE outperforms VAEs and raw pixels for both understanding and generation (Section 3). **(2) Data:** modalities minimally interfere and even exhibit positive synergy (Section 4). **(3) World Modeling:** physical prediction emerges from general pretraining, not domain-specific data (Section 5). **(4) Architecture:** MoE naturally learns modality-specific specialization (Section 6). **(5) Scaling:** vision is more data-hungry than language; MoE harmonizes this asymmetry (Section 7).

## 2. Experiment Setup

Our goal is to study the science and design choices around unified multimodal pretraining. Across experiments, we keep the compute budget and training hyperparameters controlled within each ablation family. Unless otherwise specified, we evaluate at the end of pretraining (without any instruction tuning). We provide an overview of our training and inference protocols here, and more details in Section C.

### 2.1. Training and Inference

**Training objectives.** Text is trained with standard autoregressive cross-entropy  $\mathcal{L}_{\text{LM}}$ . For visual tokens, we employ frame-wise flow matching (Lipman et al., 2023; Liu et al., 2023): given clean latents  $z_0$  and noise  $\epsilon \sim \mathcal{N}(0, I)$ , we sample  $t \sim \mathcal{U}[0, 1]$ , construct  $z_t = (1-t)\epsilon + tz_0$ , and minimize  $\mathcal{L}_{\text{flow}} = \mathbb{E}_{t, z_0, \epsilon} [\|v_\theta(z_t, t, \cdot) - (z_0 - \epsilon)\|_2^2]$ . The joint loss is  $\mathcal{L} = \lambda_{\text{LM}}\mathcal{L}_{\text{LM}} + \lambda_{\text{flow}}\mathcal{L}_{\text{flow}}$  with  $\lambda_{\text{LM}}=1.0$ ,  $\lambda_{\text{flow}}=3.0$ . We shift the noise schedule towards the noisier end (Esser et al., 2024), following RAE (Zheng et al., 2026; Tong et al., 2026). In Sections 3 and 6.1, we additionally explore  $x$ -prediction (Li & He, 2025). Training uses a hybrid attention mask via FlexAttention (Dong et al., 2024): causal for text, block-wise causal for vision (bidirectional within a frame, causal across frames).

**Inference.** Text follows autoregressive next-token prediction. Visual generation uses a 25-step Euler sampler with

CFG=3.0 (Ho & Salimans, 2022), then a pretrained decoder maps latents to pixels.

### 2.2. Model and Tokenization

**Unified decoder-only backbone.** We use a decoder-only Transformer (Vaswani et al., 2017) similar to Transfusion (Zhou et al., 2025), trained *from scratch* on a mixture of vision and language data. Unlike Transfusion, we use simple linear projection layers instead of a U-Net (Ronneberger et al., 2015). Within each self-attention block, we use modality-specific FFNs (one for text and one for vision tokens) instead of shared FFNs by default. We verify this in Figure 15: modality-specific FFNs unanimously improve performance—reducing text perplexity while improving image generation and VQA, consistent with prior findings (Lin et al., 2024). Our default model has 2.3B total parameters with 1.5B active per token. In Section 6.1, we explore learnable routing via MoE.

**Text tokens.** Text is tokenized with a standard BPE tokenizer from LLaMA-3 (Grattafiori et al., 2024) and trained with next-token prediction. We denote text tokens as  $\text{T}$ .

**Visual tokens.** An image (or video frame) is mapped to latent tokens using a *frozen* visual encoder. Our architecture is encoder-agnostic. In Section 3, we explore encoders ranging from semantic encoders (e.g., SigLIP 2 (Tschannen et al., 2025), DINOv2 (Oquab et al., 2023)) to VAEs (Kingma & Welling, 2014) to raw pixels. By default, we use SigLIP 2 So400M. We denote image tokens as  $\text{I}$ . Videos are encoded frame-by-frame at  $224 \times 224$  resolution.

### 2.3. Data Sources

We use four broad categories of training data: (i) large-scale web text (DCLM (Li et al., 2024)), (ii) raw videos (YouTube and curated video datasets (Kay et al., 2017; Goyal et al., 2017; Zellers et al., 2022)) at 1 FPS, (iii) paired image-text data (MetaCLIP (Xu et al., 2024) and in-house Shutterstock), and (iv) action-conditioned navigation trajectories formatted as  $\text{I} + \text{T} \rightarrow \text{I}$  from NWM (Bar et al., 2025) and annotated video data (see Section B.3). Examples of each data type

are shown in Figure 14.

## 2.4. Evaluation

We report: (i) **text perplexity** (PPL) on the held-out DCLM (Li et al., 2024) validation set and in-house “Notes” text corpus for more OOD evaluation, (ii) **diffusion loss** on the held-out CC12M (Changpinyo et al., 2021) validation set, (iii) **image generation score** on DPGBench (Hu et al., 2024) and GenEval (Ghosh et al., 2023), and (iv) **average VQA accuracy** on the 16 Cambrian evaluation benchmarks (Tong et al., 2024a). All VQA results are reported after 1 epoch of Cambrian-7M finetuning.

## 3. Visual Representations for Unified Multimodal Pretraining

It is well understood that semantic encoders perform better for understanding tasks such as VQA (Tong et al., 2024a). Yet, it has been widely assumed that VAE is necessary for visual generation (Liu et al., 2024c; Skorokhodov et al., 2025). Consequently, approaches like Janus (Wu et al., 2025; Ma et al., 2025), BAGEL (Deng et al., 2025) and others (AI et al., 2025; Fan et al., 2025b) use dual representations for generation and understanding, such as VAE and SigLIP 2 respectively. However, this dichotomy complicates model design and adds overhead to both training and inference.

In contrast, recent studies on Representation Autoencoders (RAE) (Zheng et al., 2026; Tong et al., 2026) demonstrate that diffusion models can operate effectively within high-dimensional latent spaces. Leveraging these findings, we simplify our design by using a *single* visual encoder for both understanding and generation.

We study three encoder families in Figure 2: VAEs (SD-VAE (Rombach et al., 2022), FLUX.1 (Labs et al., 2025)), semantic encoders (SigLIP 2 (Tschannen et al., 2025), DINOv2-L (Oquab et al., 2023), WebSSL-L (Fan et al., 2025a)), and raw pixels patchified into  $14 \times 14$  tokens. Semantic encoder latents are decoded via the off-the-shelf RAE decoder.

**Text performance.** We observe that all visual representations achieve similar (and sometimes slightly better) text perplexity compared to the text-only baseline, with raw pixels performing the best. The marginal difference indicates that multimodal pretraining does not significantly affect language capabilities regardless of visual encoder choice.

**Visual generation and understanding.** Semantic encoders consistently outperform VAEs on both visual understanding and generation. SigLIP 2 outperforms FLUX.1 on VQA and on generation benchmarks (DPGBench, GenEval), echoing RAE findings (Zheng et al., 2026; Tong et al., 2026)

that high-dimensional representations are effective for generation. A single encoder suffices for both tasks; we adopt SigLIP 2 as the default.

We also evaluate raw pixels with  $x$ -prediction (Li & He, 2025). While raw pixels underperform semantic encoders in generation, the VQA gap is narrow, suggesting direct pixel learning remains promising given sufficient compute.

## 4. Understanding the Impact of Data

We study the impact of pretraining data composition. Our unified framework (Sec. 2) is capable of modeling any combination of multimodal data. We first compare text-only pretraining against multimodal pretraining (Sec. 4.1), then dissect the role of text distribution in I/T data (Sec. 4.2), and finally demonstrate positive transfer from multimodal co-training (Sec. 4.3).

### 4.1. Pretraining Data Composition

The premise of unified multimodal pretraining is to leverage all available data. However, it is unclear how each data type contributes to or interferes with the final model. To better understand this, we study three representative data mixtures (see Section B.1 for details): **1) Text + Video** (raw video without text annotations), **2) Text + MetaCLIP** (I/T pairs) and **3) Text + Video + MetaCLIP + Action** (all the above + action-conditioned videos). All multimodal models are trained on  $\approx 1$ T tokens (520B text + 520B multimodal) and are compared to a text-only baseline trained on 520B tokens.

**Vision data minimally impacts text performance.** In Figure 3, Text + Video achieves the best perplexity on both DCLM and Notes, even outperforming the text-only baseline on DCLM—suggesting that vision itself is not the primary cause of modality competition. In contrast, Text + MetaCLIP achieves the worst perplexity, while the full mixture only slightly degrades, suggesting video and action data are complementary to text. We hypothesize that text degradation stems from distributional shifts in image captions, and investigate this in Sec. 4.2.

### I/T data enables visual generation and understanding.

Consistent with prior work (Tong et al., 2024a; Zhu et al., 2025), in Figure 4, we find that I/T data is essential for acquiring understanding and generation capabilities. Further, our results on VQA demonstrate that understanding tasks benefit significantly from broader data diversity, with the full data mixture surpassing all other data combinations.

### 4.2. Image-Text Distribution

In Section 4.1, we found that I/T data slightly degrades text perplexity. We hypothesize this stems from distribu-

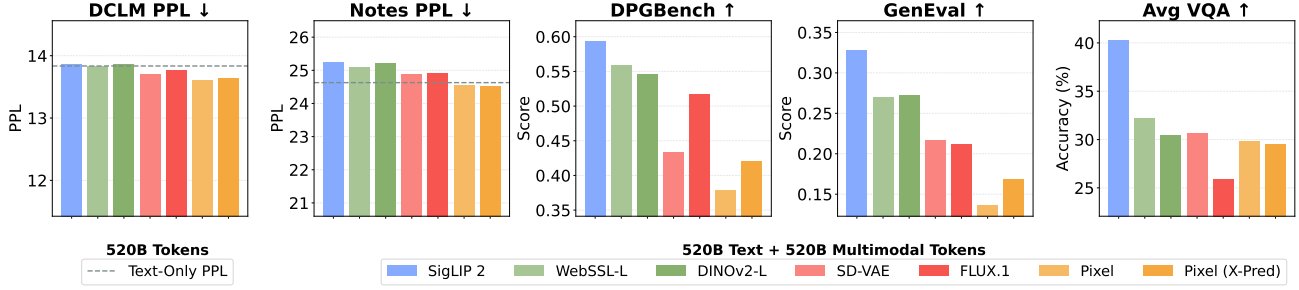


Figure 2. **RAE outperforms VAEs for both generation and understanding.** RAE (SigLIP 2) achieves the best performance on DPGBench, GenEval, and VQA while maintaining text perplexity comparable to the text-only baseline.

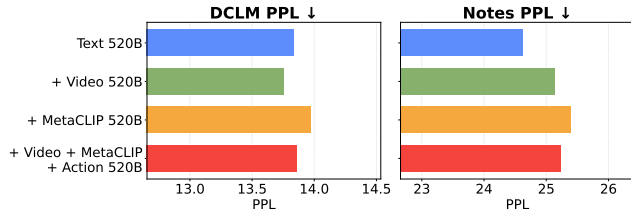


Figure 3. **Visual data does not compete with text.** Text + Video matches the text-only PPL on DCLM, suggesting that raw visual data is compatible with language modeling.

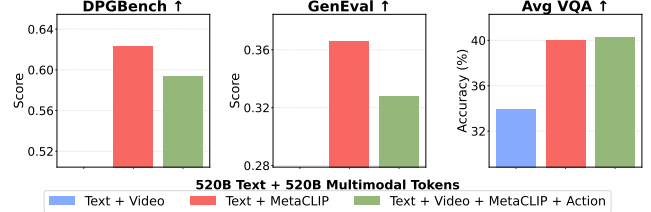


Figure 4. **I/T data enables visual capabilities.** I/T pairs are essential for visual understanding and generation. Text + Video contains no I/T data, so image generation is omitted.

tional gaps between pretraining text and image captions, and study three I/T sources: web-crawled (MetaCLIP), synthetic (MetaCLIP Recaption), and high-aesthetic (Shutterstock/SSTK). Table 3 shows that synthetic and in-house data exhibit larger cosine distance from DCLM than MetaCLIP captions, correlating with perplexity degradation (Figure 5).

Different I/T sources exhibit complementary strengths: recaptioned data improves VQA while SSTK improves generation (Figure 5). Using MetaCLIP for understanding and SSTK for generation captures the best of both, suggesting that data can be independently chosen per capability.

### 4.3. Synergy in Multimodal Pretraining

We observed in Section 4.1 that adding diverse multimodal data in equal proportion to text minimally impacts text performance. We now explore whether multimodal pretraining can actively *benefit* any capabilities.

**Vision and language are complementary.** We train models across all combinations of  $\{0, 25, 50, 75, 100\}$ B text and multimodal tokens (Figure 6). Adding text to a fixed vision budget consistently improves diffusion loss and GenEval, surpassing the multimodal-only baseline. We confirm that vision minimally impacts language (Section E.3), while language helps vision—likely because generation benchmarks are text-conditioned, so better language modeling improves text-to-image alignment.

**Synergy extends to visual understanding.** We train models on 20B VQA tokens supplemented with 80B of hetero-

geneous data (video, MetaCLIP, or text), comparing against 20B and 100B VQA-only baselines. As shown in Fig. 7, all mixed variants outperform even the 100B baseline despite using  $5\times$  less in-domain data, showing that diverse pretraining provides a stronger foundation than task-specific scaling.

We confirm this at scale: pretraining on 520B text + 520B multimodal tokens then finetuning on VQA (Tong et al., 2024a) outperforms a text-only baseline across all visual representations (Figure 19; see Section F.5). Semantic encoders such as SigLIP 2 consistently outperform VAEs, reinforcing Section 3.

## 5. Towards World Modeling in Unified Multimodal Models

We explore whether multimodal models can do world modeling (Nguyen & Widrow, 1989; Ha & Schmidhuber, 2018; LeCun, 2022) without any architectural changes. We adopt the Navigation World Model (NWM) (Bar et al., 2025) setting: predict the next visual state given context and a navigation action ( $\text{State}_t + \text{Action} \rightarrow \text{State}_{t+1}$ ).

Unlike NWM, which encodes actions as specialized continuous vectors, we represent actions directly as text tokens (i.e. numerical strings), framing the task as  $\text{I} + \text{T} \rightarrow \text{I}$  prediction (Figure 28). We introduce no action-specific adapters or architectural modifications.

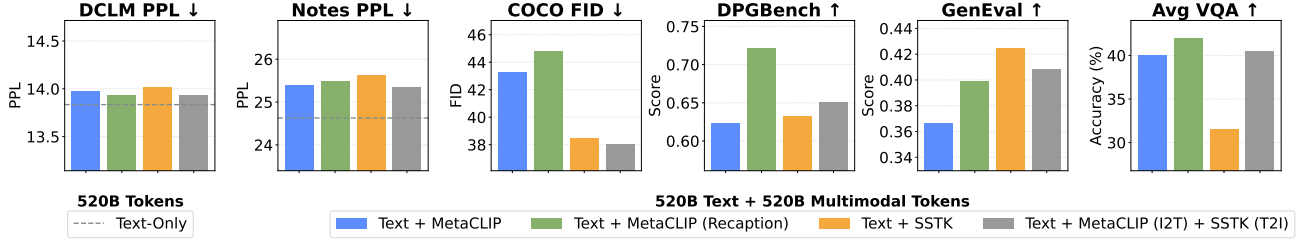


Figure 5. I/T data sources exhibit complementary trade-offs. Each source of I/T data has distinct strengths across text perplexity, generation, and VQA. Combining MetaCLIP (I2T) with Shutterstock (T2I) captures the strengths of both.

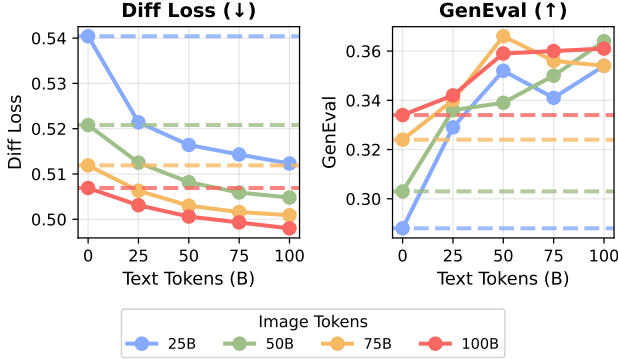


Figure 6. Multimodal co-training exceeds unimodal performance. Across all token budgets, adding text tokens to a fixed vision budget consistently improves generation quality (—), surpassing the multimodal-only baseline (—).

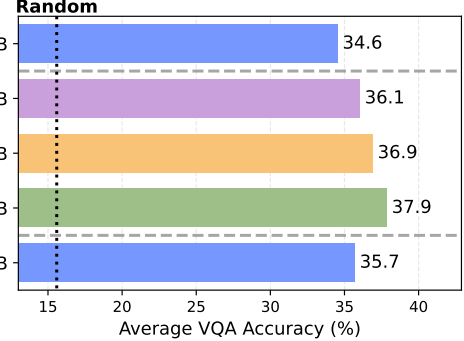


Figure 7. General pretraining outperforms specialized scaling. Supplementing 20B VQA tokens with general-purpose data (text, video, or image-text) yields higher accuracy than training on 100B VQA tokens alone.

### 5.1. NWM Data and Protocol

We follow NWM (Bar et al., 2025), training on egocentric navigation datasets (SCAND, RECON) (Shah et al., 2021; Karnan et al., 2022) where each sample has 4 context frames, an action, and a target frame. We evaluate via zero-shot CEM planning (Rubinstein, 1997) with  $N=120$  samples over an 8-step horizon, reporting absolute trajectory error (ATE) and relative pose error (RPE). See Sec. I.1 for details.

### 5.2. World Modeling Emerges from Multimodal Pretraining

Does world modeling stem from domain-specific data, or from broader multimodal capabilities? We compare models trained on 50B NWM tokens + 50B multimodal data against a 50B NWM-only baseline.

As shown in Fig. 8, multimodal pretraining outperforms NWM-only scaling (50B→100B), with pure video yielding the biggest gain. All other modalities also help, suggesting that world modeling relies more on general multimodal capabilities than domain-specific data (Hafner et al., 2025).

**World modeling transfers from general training.** We further ablate the ratio of NWM data while keeping the total budget fixed at 200B tokens (Section I.1). As shown in Fig. 9, performance saturates at just 1% in-domain data,

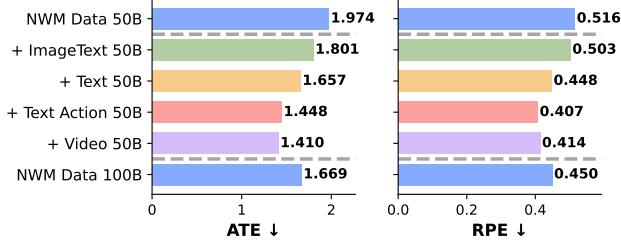
reinforcing that capabilities such as navigation and VQA are primarily acquired during general pretraining. Beyond predefined actions, our model enables *zero-shot* conditioning on free-form language prompts (e.g. “get out of the shadow!”), with generated frames maintaining visual consistency while faithfully executing commanded actions; see Section I.2 for qualitative examples.

## 6. Unified Multimodal Architecture Design

We show in Fig. 15 that modality-specific FFNs improve over shared FFNs, but this evenly splits capacity and may not reflect ideal allocation. We explore whether MoE can learn this separation dynamically. We study the MoE design space (Sec. 6.1), emergent specialization (Sec. 6.2), and progressive stacking of design choices (Sec. 6.3).

### 6.1. Exploring MoEs for Unified Multimodal Models

MoE generalizes fixed modality-specific FFNs by routing each token to a subset of specialized experts (Shazeer et al., 2017; Fedus et al., 2022), with demonstrated scalability for language (Liu et al., 2024a;b; OpenAI, 2025; Team et al., 2025; Yang et al., 2025). Here, we study its design for multimodal pretraining. Unless noted, all MoE experiments train for 57B tokens on a 50/50 DCLM text / Shutterstock image mix.



**Figure 8. Unsupervised video data drives world modeling performance.** Supplementing the NWM baseline with general-purpose video data yields the best performance.

**Granularity.** We first analyze the role of expert *granularity*  $G$ , defined relative to model dimension  $d_{\text{model}}$  and expert dimension  $d_{\text{expert}}$  as  $G = 4d_{\text{model}}/d_{\text{expert}}$ . We sweep  $G \in \{1, 4, 16, 32, 64\}$  at fixed active compute: as  $G$  increases, the number of active experts increases while the dimension of each expert shrinks proportionally.  $G=1$  corresponds to a baseline of 16 large experts ( $d=8192$ ) with Top-1 routing, while  $G=64$  yields 1024 small experts ( $d=128$ ) with Top-64 routing.

We test two visual representations—RAE (SigLIP 2) and VAE (FLUX.1)—to understand how the visual latent space affects MoE efficacy. We also test two diffusion prediction targets:  $x$ -pred (Li & He, 2025) and  $v$ -pred (Lipman et al., 2023). All models have 13.5B total parameters (1.5B active); see Section F.6 for detailed configurations.

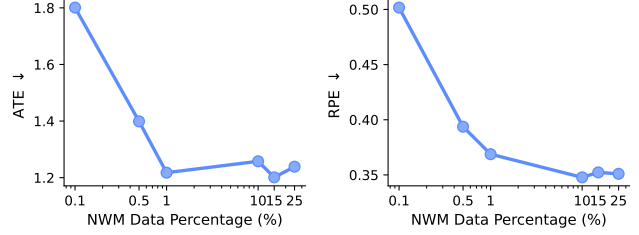
As shown in Figure 20, **higher granularity is critical**: increasing  $G$  from 1 to 16 substantially improves both modalities, as many smaller experts enable better capacity allocation. Notably, vision saturates at  $G = 4$  while language at  $G = 16$ , suggesting language benefits more from fine-grained routing. Gains plateau beyond  $G = 32$ ; we adopt  $G = 16$ .

#### Prediction target depends on visual representation.

The optimal diffusion prediction target varies with the visual representation (Figure 20, solid vs. dashed). For RAE (SigLIP 2),  $x$ -pred consistently outperforms  $v$ -pred on image generation across all granularities, with minimal impact on language. For VAE (FLUX.1),  $x$ -pred causes text perplexity to spike as granularity increases, while  $v$ -pred scales cleanly. This suggests that  $x$ -pred is less stable for low-dimensional representations as expert dimension shrinks, while high-dimensional representations benefit.

**Sparsity.** Using  $G = 16$ , we fix the active budget (16 experts, hidden dim 512) and increase the total expert count from 32 to 1008, reducing the active ratio from 50% to 1.6%. We compare against a 1.5B dense baseline at same compute.

As shown in Figure 21, both modalities benefit consistently from increased sparsity: text PPL decreases and visual gen-



**Figure 9. World modeling transfers with minimal alignment.** Performance saturates at 1%, suggesting that the core capability is mostly acquired from general pretraining.

**Table 1. Per-Modality Shared Experts outperform Global Shared Experts.**

| Config    | DCLM<br>PPL ↓ | Notes<br>PPL ↓ | Diff.<br>Loss ↓ | Gen-<br>Eval ↑ |
|-----------|---------------|----------------|-----------------|----------------|
| No Shared | 14.80         | 27.39          | 0.484           | 0.360          |
| Global    | 14.79         | 27.25          | 0.483           | 0.364          |
| Per-Modal | <b>14.79</b>  | <b>27.16</b>   | <b>0.483</b>    | <b>0.367</b>   |

eration improves as total experts grow from 32 to over 1000—all at fixed compute. This confirms that MoE extends naturally from language to unified multimodal pretraining.

The training loss (Figure 24) reveals a further distinction: for RAE (SigLIP 2), both text and diffusion loss continue improving with more experts, while for VAE (FLUX.1) diffusion loss saturates, suggesting that semantic representations scale better with sparsity.

**Expert design choices.** Following Dai et al. (2024); Liu et al. (2024a), we test whether always-active “shared experts” help. We compare three configurations (256 total, 16 active): (1) **No Shared**: standard top-16. (2) **Global Shared**: 1 always-active expert; top-15 from 255. (3) **Per-Modality Shared**: 1 shared expert per modality; top-15 from 254.

As shown in Table 1, per-modality shared experts yield the best performance across all metrics, suggesting different modalities have distinct capacity needs—a theme we examine in Section 6.2.

## 6.2. Emergent Expert Specialization

Having established that MoE is effective for multimodal pretraining, we analyze what the model learns. We examine a 13.5B-parameter MoE model (1.5B active,  $G=16$ , 256 experts, SigLIP 2,  $x$ -pred) trained on 1T tokens (see Section F.7 for methodology).

**Modality specialization emerges naturally.** We classify each expert as Text, Vision, or Multimodal based on routing preferences across held-out data. As shown in Figure 22, the model allocates more experts to text than vision de-

spite load-balancing—consistent with our scaling laws (Section 7): language is parameter-hungry while vision is data-hungry (Wang et al., 2026). Expert composition evolves across depth: early layers are text-dominated while later layers contain more vision and multimodal experts, suggesting a separate-then-integrate strategy.

**Vision experts are general-purpose.** We measure expert selection rates across 10 diffusion timestep bins. CV scores cluster near 0.15 (Figure 23), indicating minimal timestep specialization. Furthermore, routing patterns for understanding vs. generation show high correlation ( $r \geq 0.90$ ) across all layers (Figure 25), suggesting a unified visual representation (Tong et al., 2025; Li et al., 2025c; Yue et al., 2026).

### 6.3. Stacking Design Choices

We now stack the best choices from each axis progressively, building from a Transfusion baseline toward our final configuration, and compare against Mixture of Transformers (MoT) (Liang et al., 2024). All models use identical hyperparameters, data, and FLOPs. Fig. 26 summarizes results.

**FFN structure.** Replacing the shared FFN with modality-specific FFNs improves perplexity from 15.93 to 15.13 and DPG from 0.45 to 0.47 (Fig. 15). This simplest capacity separation reduces modality competition at no extra inference cost, since only one FFN is active per token (see Sec. D).

**Vision encoder.** Using modality-specific FFN, we compare four vision encoders: three single-encoder configurations (SD-VAE, FLUX.1, and SigLIP 2 for RAE-style diffusion) and a dual-encoder setup that uses SigLIP 2 for understanding and SD-VAE for generation, as in Bagel (Deng et al., 2025) and Janus (Ma et al., 2025; Chen et al., 2025b). SigLIP 2 significantly outperforms all alternatives, improving DPG from 0.47 to 0.57 while also reducing perplexity from 15.13 to 15.06. The dual-encoder baseline falls short of single SigLIP 2 on both metrics.

**Parameter separation strategy.** Using SigLIP 2 and modality-specific FFN, we compare dense (no further separation), deep separation via MoT (Liang et al., 2024; Deng et al., 2025), and data-driven separation via MoE. MoE achieves the best trade-off, reaching the lowest perplexity (12.49) and highest DPG score (0.63), outperforming both the dense baseline and MoT.

MoE also has a structural advantage: learned routing adapts capacity per token, whereas fixed separation cannot (Sutton, 2019). MoT has proven effective when finetuning pretrained dense LLMs (Liang et al., 2024; Deng et al., 2025); as language models adopt native MoE (Liu et al., 2024b; Team et al., 2025), extending routing with vision experts is a natural next step.

**Prediction target.** Within the best architecture (MoE + SigLIP 2), switching from  $v$ -pred to  $x$ -pred improves DPG from 0.63 to 0.65.

**Knowledge-informed generation.** We also evaluate on WISE (Niu et al., 2025), which probes factual understanding across categories such as science and culture (Fig. 27). Semantic encoders outperform VAEs by 3–4 $\times$  across all categories, and MoE achieves the highest score, confirming that learned routing best transfers semantic knowledge.

## 7. Scaling Laws for Unified Multimodal Models

Prior scaling laws focus on unimodal language (Kaplan et al., 2020; Hoffmann et al., 2022; Tian et al., 2026) or text-only multimodal output (Shukor et al., 2025). We derive laws for both modalities and study how architecture influences these trends.

**IsoFLOP methodology.** Following Hoffmann et al. (2022), we sweep model sizes  $N$  and token counts  $D$  at fixed compute budgets  $C$  (estimated via `torch titan` (Liang et al., 2025);  $6ND \approx C$  for dense; active parameters for MoE). We fit parabolas in log-parameter space to IsoFLOP curves on held-out validation loss (DCLM for language, CC12M for vision), then extract power-law relationships  $N_{\text{opt}} \propto C^a$  and  $D_{\text{opt}} \propto C^b$  where  $a + b = 1$ .

**Compute-optimal for dense models.** Figure 10 presents dense IsoFLOP results. For **Language**,  $a \approx 0.47$  and  $b \approx 0.53$ , nearly balanced and consistent with Chinchilla. For **Vision**,  $a \approx 0.37$  and  $b \approx 0.63$ , indicating vision is more data-hungry. No single compute-optimal trend governs both modalities. Relating optimal tokens to model size ( $D_{\text{opt}} \propto N_{\text{opt}}^{b/a}$ ), the ratio of required vision to language data grows as  $\mathcal{O}(N^{0.57})$ : from a 1B baseline, demand grows by 14 $\times$  at 100B and 51 $\times$  at 1T parameters.

**Compute efficiency for dense models.** We fit  $L(C) = A \cdot C^{-\alpha} + E$  (Kaplan et al., 2020) to Dense Multimodal, Text-Only, and T2I-Only baselines. Fig. 11 shows the unified model matches or exceeds both: for **Language**, DCLM PPL 13.3 vs. 13.6 at  $10^{21}$  FLOPs; for **Vision**, DPG 0.622 vs. 0.598 and FID 39.3 vs. 41.5, suggesting joint training benefits visual generation quality.

**Compute-optimal for MoE models.** Using sparsity ratio 16 (Section 6.1), Figure 12 presents MoE IsoFLOP results. **Language** exhibits  $a \approx 0.41$ ,  $b \approx 0.59$ , while **Vision** exhibits  $a \approx 0.36$ ,  $b \approx 0.64$ —halving the exponent gap from 0.10 (dense) to 0.05. The language data exponent  $b$  shifts from 0.53 to 0.59, moving closer to vision’s data-intensive

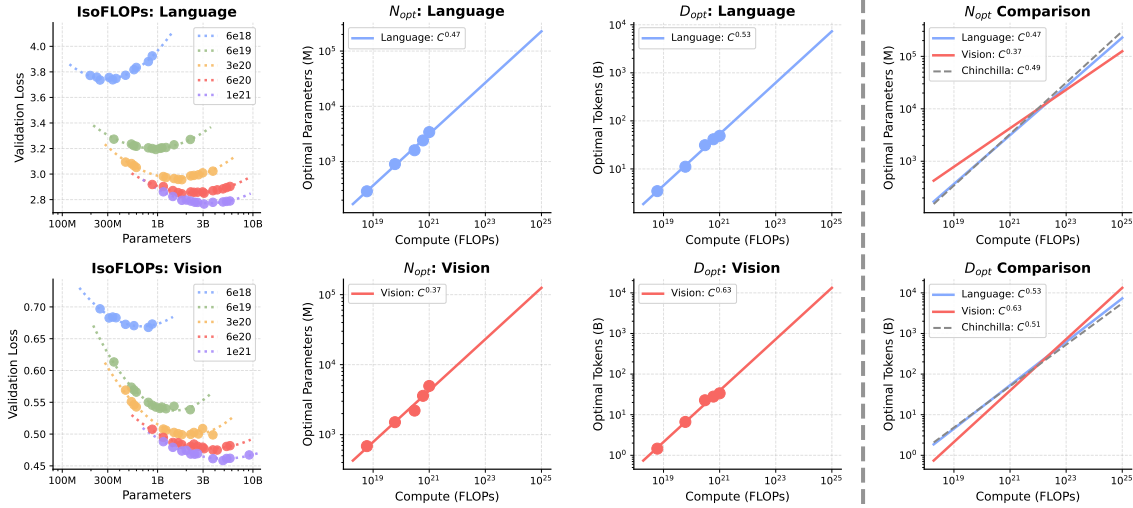


Figure 10. **Scaling laws for unified dense models.** We compute IsoFLOP curves across compute budgets. We project the optimal model size and token count for each FLOP budget, and find that the vision modality is more data-hungry, while text follows Chinchilla-like scaling.

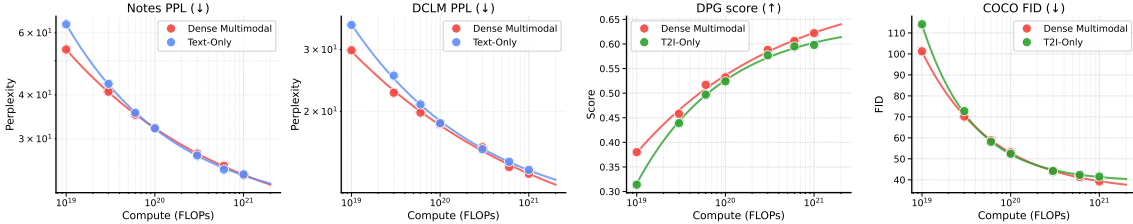


Figure 11. **Scaling laws and efficiency benchmarks for unified dense models.** We fit  $L(C) = A \cdot C^{-\alpha} + E$  to each model class. Dense Multimodal outperforms both the unimodal Text-Only and T2I-Only baselines across all four metrics.

regime, making sparsity a practical lever to harmonize divergent demands.

**MoE compute efficiency.** Fig. 13 compares MoE Multimodal against unimodal MoE baselines. Across all four metrics, the unified model closely tracks its unimodal counterparts (DCLM PPL 12.3 vs. 12.0; FID 39.2 vs. 39.8 at  $10^{21}$  FLOPs), confirming that a single model matches unimodal performance while leveraging model sparsity.

## 8. Related Work

**Multimodal pretraining and unified models.** Early unified methods used discrete tokens (Oord et al., 2017; Razavi et al., 2019; Lee et al., 2022) for next-token prediction (Lu et al., 2022a; Aghajanyan et al., 2022; Team, 2024; Wang et al., 2024), while recent work incorporates diffusion (Zhou et al., 2025; Cui et al., 2025). Current models (Zhang et al., 2025a; Xiao et al., 2025; Geng et al., 2025; Li et al., 2025a; Nguyen et al., 2025; Li et al., 2025b; Zhang et al., 2026) vary in architectural separation, from adapter-based (Dai et al., 2023; Ge et al., 2024; Tong et al., 2025) to joint training (Liang et al., 2024; Shi et al., 2024; Ma et al., 2025;

Liao et al., 2025; Deng et al., 2025; Xie et al., 2025). We build on Transfusion and compare these in Section 6.3.

**Visual representations.** Understanding uses semantic latents from supervised (Radford et al., 2021; Tschannen et al., 2025) or self-supervised (He et al., 2021; Oquab et al., 2023; Fan et al., 2025a) encoders, while generation favors VAE latents (Kingma & Welling, 2014; Rombach et al., 2022; Labs, 2024). Recent work (Chen et al., 2025a; Zheng et al., 2026; Tong et al., 2026) shows semantic latents also enable strong generation. We show that a single RAE encoder excels at both tasks.

**World models and scaling laws.** World models (Nguyen & Widrow, 1989; Ha & Schmidhuber, 2018; LeCun, 2022) evolved from in-domain (Hafner et al., 2020, 2023; Hansen et al., 2024; Hafner et al., 2025) to video generators (Bruce et al., 2024; Peng et al., 2025; Wiedemer et al., 2025). We test whether world modeling emerges from multimodal pretraining (Section 5). For scaling, Chinchilla (Hoffmann et al., 2022) quantified compute-optimal tradeoffs, extended to MoE (Tian et al., 2026) and diffusion (Polyak et al., 2024). We derive joint vision-language laws revealing an

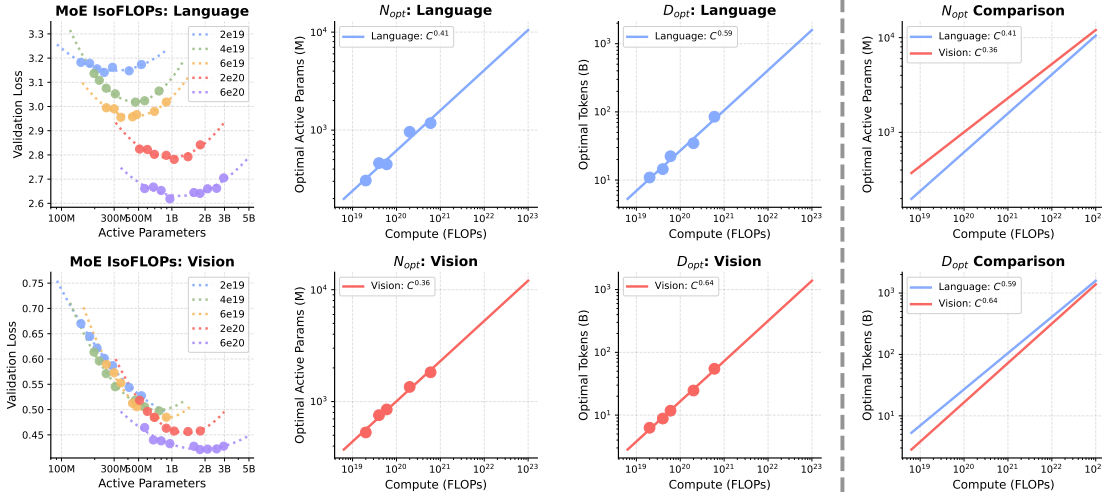


Figure 12. **Scaling laws for multimodal MoE models.** MoE narrows the vision-language exponent gap from 0.10 (dense) to 0.05 ( $D_{opt} \propto C^{0.64}$  for vision vs.  $C^{0.59}$  for language).

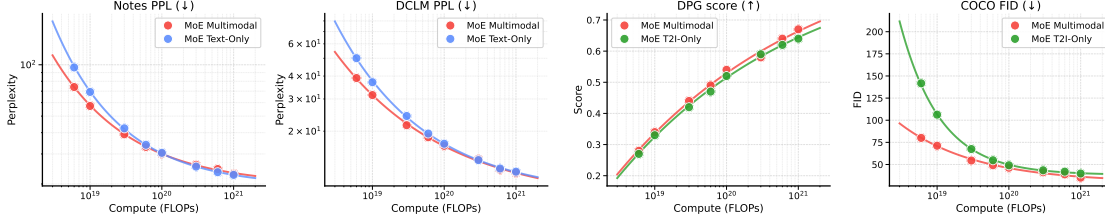


Figure 13. **Scaling laws and efficiency benchmarks for unified MoE models.** MoE Multimodal closely tracks unimodal MoE baselines across all four metrics, with near-identical FID and competitive language perplexity at  $10^{21}$  FLOPs.

asymmetry that MoE harmonizes (Section 7).

## 9. Discussion

By training unified multimodal models from scratch and systematically varying one factor at a time, we find that modality competition is not a fatal flaw—it is a symptom of specific design choices.

**A single visual representation suffices.** Contrary to prior works that use separate representations (Deng et al., 2025; Wu et al., 2025), we join a growing body of work (Shi et al., 2025; Chen et al., 2025a; Tong et al., 2026; Zheng et al., 2026; Shi et al., 2026) showing a single semantic representation (RAE) excels at both understanding and generation (Section 3), with MoE analysis confirming shared expert activation (Section 6.2). RAE also continues improving with capacity while VAE methods saturate (Figure 24).

**Modality competition is largely solvable.** The “modality tax” has two identifiable sources (Lin et al., 2024): distributional shifts in image captions (Section 4), not vision itself, and rigid capacity allocation in dense models, which MoE resolves via learned routing (Section 6.1). Pure video complements language, and co-training yields positive transfer for VQA and world modeling (Secs. 4.3, 5).

**Efficient and effective scaling with MoE.** MoE has proven effective for language (Liu et al., 2024a;b), but multimodal pretraining requires additional design (Sec. 6.1). Our IsoFLOP analysis (Sec. 7) reveals a scaling asymmetry—language follows Chinchilla-like balanced scaling while vision is more data-hungry—that MoE narrows by shifting language toward a data-hungry regime. Sparsity helps harmonize modalities with distinct scaling behaviors.

**What’s next?** The finding that world modeling capabilities emerge from general multimodal pretraining with minimal domain-specific data (Sec. 5) aligns with the vision that general-purpose models, trained broadly on how the world looks and moves, can develop an internal model of physical reality (LeCun, 2022). The ingredients are already available: vast quantities of unlabeled video remain largely untapped, and our results show they can be incorporated without degrading language. As unified models grow in capacity and data, the boundary between multimodal and world models will blur, and future models will evolve into truly native, end-to-end multimodal systems. We hope our insights encourage extending the foundation model paradigm beyond LLMs, toward systems that understand and reason about the physical world.

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## Impact Statement

Our work advances the scientific understanding of unified multimodal pretraining. While we recognize that multimodal models have many potential use cases across a large number of application areas that may have societal impacts, our work is only focused on achieving better scientific understanding. The societal impact of AI deserves continued consideration and discussion from the larger community.

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## A. Limitations and Future Work

We focus on investigating multimodal pretraining from scratch rather than post-training. Unified multimodal models can improve greatly through RL or even multimodal RL by enabling the system to both generate and interpret visual latents. On the architecture front, while we explore many facets of multimodal Mixture-of-Experts (MoE) design, our scope is not exhaustive; for instance, the uneven distribution of tokens per expert remains a significant bottleneck for hardware utilization. Conversely, leveraging higher granularity may enhance the scaling efficiency of these MoEs. Regarding data, our work excludes interleaved data, which remains a critical area for future exploration. In the long run, we believe the capability to spontaneously generate visual latents will drive the convergence of multimodal models and world modeling, fostering “System 2” behaviors that are grounded in the real world.

## B. Data

This section provides details on the training data used throughout the paper (Sec. 2.3). Figure 14 shows examples of each data type.

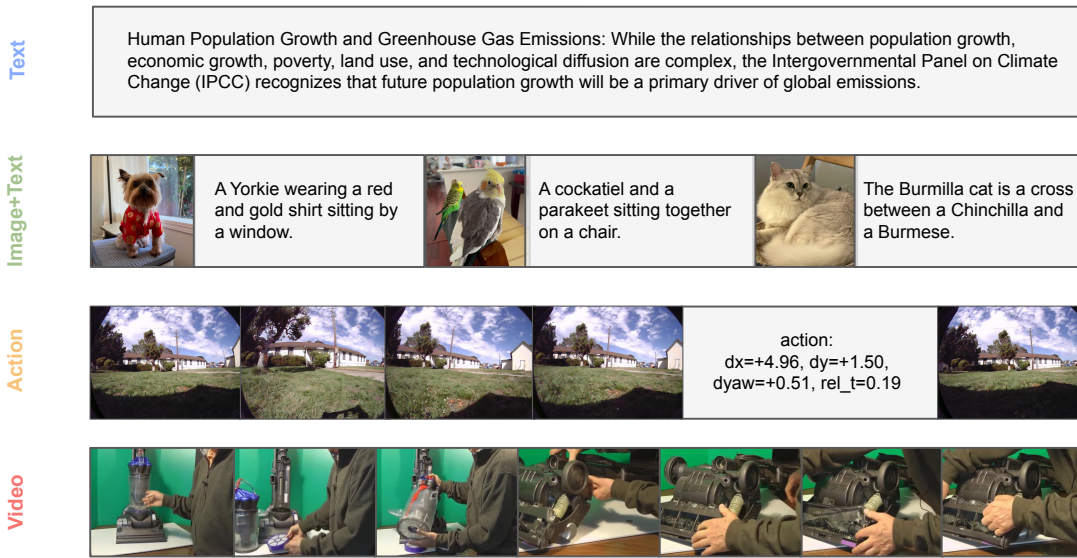


Figure 14. **Examples of training data.** We train on diverse data including text, videos, image-text pairs, and action-conditioned video.

### B.1. Data Sources

Table 2 summarizes the data sources and supervision signals used in pretraining (Sec. 2.3).

| Data Type          | Modality                           | Source                                                                                                              |
|--------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Text               | $T \rightarrow T$                  | DCLM (Li et al., 2024)                                                                                              |
| Video              | $I \rightarrow I$                  | YouTube-Temporal 1B (Zellers et al., 2022), SomethingSomething V2 (Goyal et al., 2017), Kinetics (Kay et al., 2017) |
| Image-Text Pairs   | $I \rightarrow T, T \rightarrow I$ | MetaCLIP (Xu et al., 2024), in-house Shutterstock                                                                   |
| Action-Conditioned | $I + T \rightarrow I$              | NWM (Bar et al., 2025), in-house Text-Annotated Videos                                                              |

Table 2. Training data sources and supervision signals.

### B.2. MetaCLIP Recaption

We recaption images using Qwen2.5-VL-32B-Instruct (Bai et al., 2025) with nucleus sampling (temperature 0.6, top-p 0.95), generating up to 128 tokens per caption.

### B.3. Video Action Annotations

We collected about 12M video + text action annotations from YouTube-Temporal 1B (Zellers et al., 2022) using a model-based pipeline, where an MLLM captions the transition between two consecutive video chunks, focusing on physical actions. The annotations are verified via self-reflection.

### B.4. Cosine Distance Computation

We sample 100K samples from each dataset, truncate at 2000 characters, extract embeddings using Qwen3-Embedding-8B (Zhang et al., 2025b), and compute the cosine distance between centroids (Table 3; see Sec. 4.2).

Table 3. Cosine distance of image captions to DCLM.

|                               | MetaCLIP | MetaCLIP (Recap) | SSTK  |
|-------------------------------|----------|------------------|-------|
| Cosine Dist. ( $\downarrow$ ) | 0.196    | 0.286            | 0.215 |

## C. Extended Methodology: Unified Multimodal Mechanics

We provide more detailed explanations of our unified pretraining and inference mechanics.

**Sequence formatting.** Text tokens are mapped to vectors via a standard `nn.Embedding` layer. Visual inputs are mapped to latent patches using a frozen visual encoder and flattened into 1D sequences. We enclose each image or video frame with beginning-of-image (`<BOI>`) and end-of-image (`<EOI>`) marker tokens. A video is processed as a series of individually bounded frames.

**Hybrid attention masking.** We implement a hybrid attention mask using FlexAttention. Text tokens apply a standard causal mask. For visual data, we apply a block-wise causal mask: patches within the same image attend bidirectionally, but can only attend causally to text or visual frames that appeared previously.

**Joint training and flow matching.** During training, we apply distinct objectives to each modality. For text, we use standard next-token prediction. For visual data, we employ frame-wise flow matching with an independent timestep  $t \sim \mathcal{U}[0, 1]$  per frame. Text tokens appearing after an image condition on the noisy visual representations.

**Loss computation and routing.** The model optimizes  $\mathcal{L} = \lambda_{LM}\mathcal{L}_{LM} + \lambda_{flow}\mathcal{L}_{flow}$  in a single forward-backward pass. Output hidden states route to modality-specific heads: vocabulary head for text cross-entropy loss, continuous linear head for flow matching loss.

**Modal-switching inference algorithm.** At inference, the model dynamically transitions between discrete autoregressive generation and continuous flow matching. Upon sampling a `<BOI>` token, the decoding suspends text generation and appends pure noise vectors. The model denoises these using a 25-step Euler sampler, then appends `<EOI>` and resumes text generation.

## D. Shared vs. Modality-Specific FFNs

Recent work demonstrates that “modality separation” improves cross-modal learning, but differs in the degree of architectural decoupling (Lin et al., 2024; Liang et al., 2024; Shi et al., 2024; Deng et al., 2025).

We study the simplest instantiation: shared self-attention with separate vision-only and language-only FFNs (“modality-specific FFN”), compared to a shared FFN per layer. While modality-specific FFN increases total parameter count, training and inference FLOPs remain constant as only one FFN is activated per token. Figure 15 demonstrates that modality-specific FFN reduces text perplexity while simultaneously improving performance in image generation and visual understanding. Consequently, we adopt the modality-specific FFN as the default.

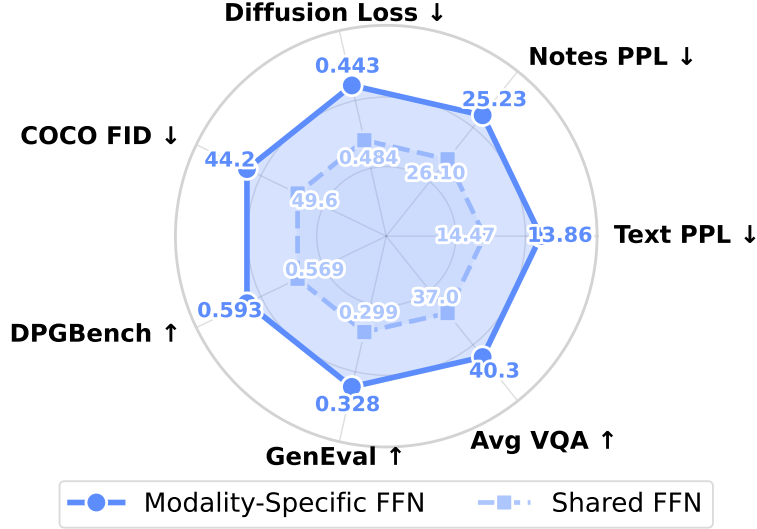


Figure 15. Shared → Modality-specific FFN. Modality separation improves multimodal learning over shared FFN.

## E. Additional Results and Exploration

This section provides additional analyses supporting the visual representation study in Sec. 3, including layer-wise encoder probing (Figure 16), loss centering, and text performance under different data compositions.

### E.1. Layer-wise Performance

Recent work shows that visual features exhibit different properties throughout different layers of the vision encoder (Bolya et al., 2025). We use layer-wise linear probing (Figure 17) to study two questions: (1) how semantic understanding and low-level pixel fidelity evolve across layers of the SigLIP 2 encoder, and (2) whether the transformer backbone preserves the encoder’s semantic representations after training.

**Representation encoder.** Across SigLIP 2 layers, ImageNet accuracy rises from 12.8% (layer 0) to 86.3% (layer 26), while PSNR drops from 29.6 dB to 20.9 dB. This reveals a fundamental trade-off where deeper encoder layers progressively discard fine-grained spatial information in favor of abstract semantic features.

**Transformer backbone.** Despite the complexity of the joint training mixture, the learned transformer preserves the semantic quality of the features. Linear probe accuracy slightly surpasses that of the frozen SigLIP input, exhibiting minor improvements across layers.

### E.2. Loss Centering

In the standard Transfusion framework, the joint loss uses fixed scalar weights throughout training. However, different visual encoders produce flow matching losses of vastly different magnitudes. Inspired by the centering mechanism in DINO (Caron et al., 2021), we replace the fixed loss weights with adaptive, per-step weights derived from an exponential moving average (EMA) of the loss magnitudes. At each training step:

$$c_{\text{current}} = \alpha \mathcal{L}_{\text{flow}} + (1 - \alpha) \mathcal{L}_{\text{LM}}, \quad (1)$$

where  $\alpha \in [0, 1]$  controls the desired emphasis. The running center is updated via EMA:  $c_{\text{target}} \leftarrow \mu c_{\text{target}} + (1 - \mu) c_{\text{current}}$ . The per-modality weights are then:

$$w_{\text{flow}} = \frac{c_{\text{target}}}{\mathcal{L}_{\text{flow}}}, \quad w_{\text{LM}} = \frac{c_{\text{target}}}{\mathcal{L}_{\text{LM}}}. \quad (2)$$

Table 4 shows results across visual encoders.

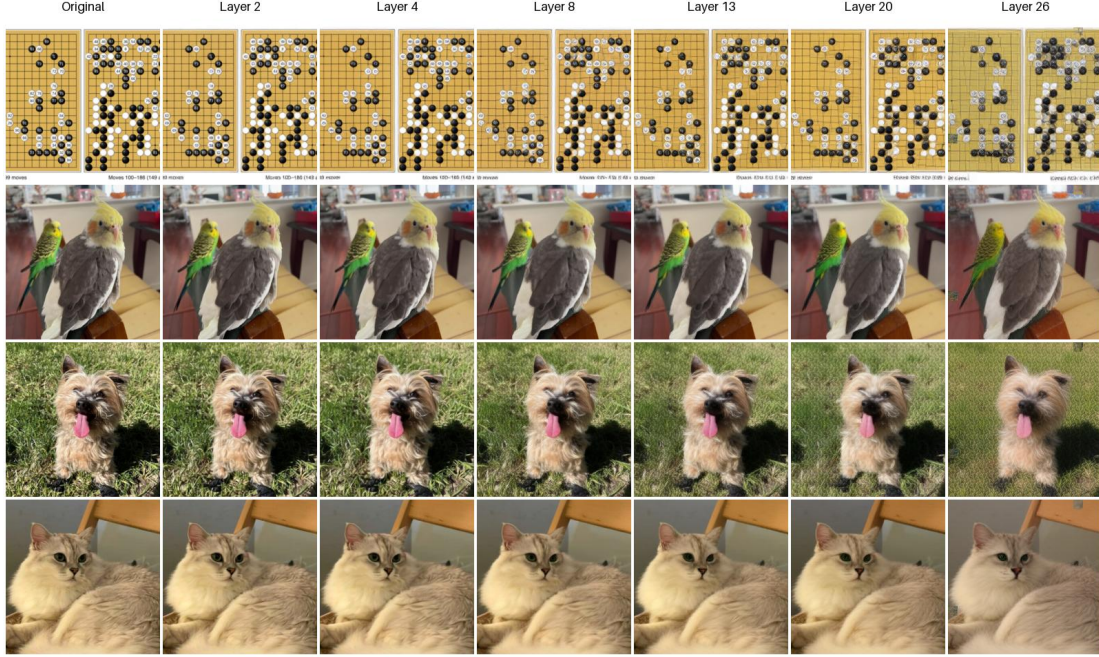


Figure 16. Qualitative reconstruction results across various SigLIP 2 layers using RAE. Earlier layers yield higher-fidelity reconstructions.

Table 4. Effect of loss centering on generation quality (DPG  $\uparrow$ ) and text perplexity (PPL  $\downarrow$ ) across visual encoders.

| Encoder  | DPG Score ( $\uparrow$ ) |                      | Perplexity ( $\downarrow$ ) |                     |
|----------|--------------------------|----------------------|-----------------------------|---------------------|
|          | Uncentered               | Centered             | Uncentered                  | Centered            |
| SigLIP 2 | 0.570                    | <b>0.612 (+.042)</b> | 15.06                       | 15.36 <b>(+.30)</b> |
| FLUX.1   | 0.498                    | 0.510 <b>(+.012)</b> | <b>14.96</b>                | 15.14 <b>(+.18)</b> |
| SD-VAE   | 0.470                    | 0.480 <b>(+.010)</b> | 15.13                       | 15.35 <b>(+.22)</b> |

### E.3. Text Performance on Data Composition Studies

Figure 18 extends the synergy analysis from Sec. 4.3 to text metrics. We measure how each modality affects the other by training models across all combinations of  $\{0, 25, 50, 75, 100\}$ B text and multimodal tokens. Consistent with Section 4.1, adding visual tokens to a fixed text budget minimally impacts text perplexity with small improvements on DCLM and slight degradation on Notes.

## F. Implementation Details

### F.1. Architecture Details

Unlike Transfusion (Zhou et al., 2025), we use simple linear projection layers instead of U-Net for visual input/output projections. Our architecture is encoder-agnostic and accepts any frozen visual encoder. For the loss weighting, we use  $\lambda_{\text{LM}}=1.0$  and  $\lambda_{\text{flow}}=3.0$  by default.

### F.2. Pretraining Details

By default, training uses a sequence length of 4096, 128 GPUs, and a batch size of 4 per GPU, for a total batch size of  $\approx 2\text{M}$  tokens per step. We use AdamW optimizer with a peak LR of  $3 \times 10^{-4}$ , 1000 warmup steps, and cosine decay to 5% of the peak LR.

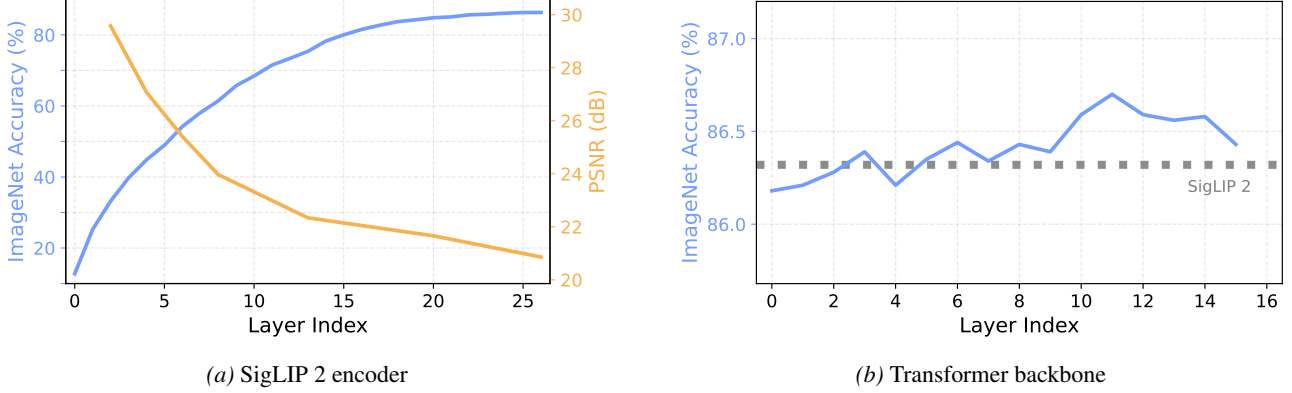


Figure 17. Layer-wise linear probing results. (a) ImageNet accuracy and PSNR across SigLIP 2 encoder layers reveal an inverse relationship between semantic understanding and pixel fidelity. (b) The Transformer backbone maintains high ImageNet accuracy throughout all layers.

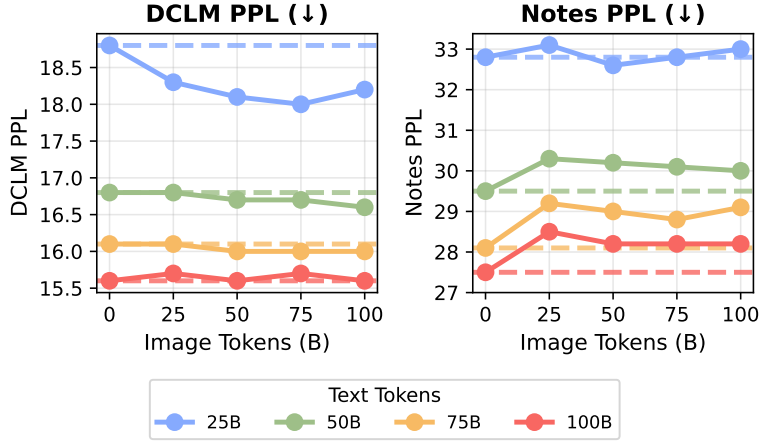


Figure 18. **Multimodal co-training exceeds unimodal performance.** We compare multimodal models (solid lines) against unimodal baselines (dashed lines) across varying budgets.

### F.3. Inference Details

Text generation uses autoregressive next-token prediction. Visual generation denoises via a 25-step Euler sampler with classifier-free guidance scale of 3.0 and 10% conditioning dropout during training (Ho & Salimans, 2022).

### F.4. Vision Encoders

Table 5 lists the encoder configurations used in the visual representation study (Sec. 3).

### F.5. VQA Evaluation

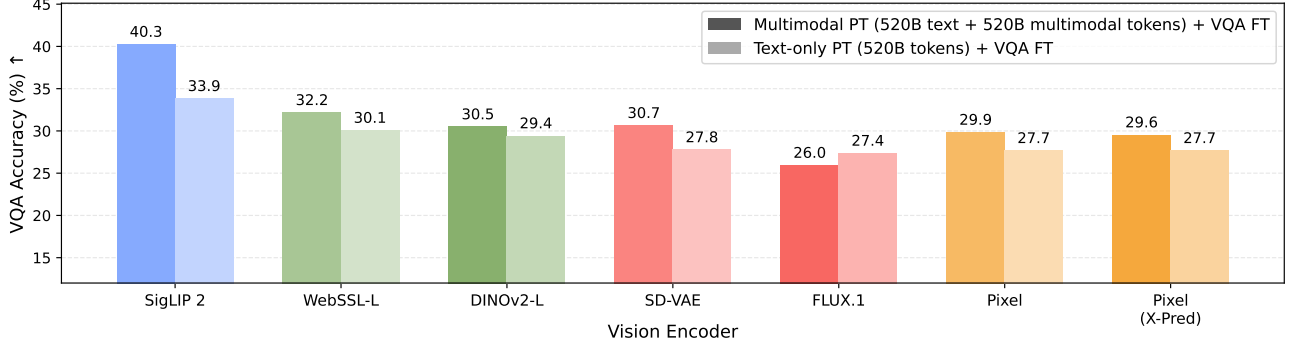
Figure 19 compares multimodal pretraining against text-only baselines on VQA (Sec. 3).

The encoder remains frozen. We finetune the multimodal model for 1 epoch on Cambrian-7M with 64 GPUs and a peak LR of  $1 \times 10^{-5}$ . For text-only models, we first warm up the vision adapter on the Cambrian-Alignment dataset and keep all other parameters frozen, using a peak LR of  $2 \times 10^{-3}$ .

We evaluate on the 16 benchmarks grouped in Tong et al. (2024a): MMBench (Liu et al., 2024d), Seed (Ge et al., 2023), MME (Fu et al., 2025), GQA (Hudson & Manning, 2019), AI2D (Hiippala et al., 2021), MathVista (Lu et al., 2023), MMMU (Yue et al., 2024), ScienceQA (Lu et al., 2022b), OCRBench (Liu et al., 2024e), TextVQA (Singh et al., 2019), ChartQA (Masry et al., 2022), DocVQA (Mathew et al., 2021), MMVP (Tong et al., 2024b), CV-Bench 2D (Tong et al.,

Table 5. Vision encoder configurations. All encoders produce 256 tokens ( $16 \times 16$  grid) per image.

| Type | Encoder   | Res.    | Patch          | Grid           | Adapt.         | Tok. |
|------|-----------|---------|----------------|----------------|----------------|------|
| Raw  | Raw Pixel | $224^2$ | $14 \times 14$ | $16 \times 16$ | —              | 256  |
| RAE  | SigLIP 2  | $224^2$ | $14 \times 14$ | $16 \times 16$ | —              | 256  |
|      | DINOv2-L  | $224^2$ | $14 \times 14$ | $16 \times 16$ | —              | 256  |
|      | WebSSL-L  | $224^2$ | $14 \times 14$ | $16 \times 16$ | —              | 256  |
| VAE  | SD-VAE    | $256^2$ | $8 \times 8$   | $32 \times 32$ | Pix[Un]shuf(2) | 256  |
|      | FLUX.1    | $256^2$ | $8 \times 8$   | $32 \times 32$ | Pix[Un]shuf(2) | 256  |


 Figure 19. **Multimodal pretraining is better for VQA.** Multimodal pretraining (dark bars) consistently outperforms text-only baselines (light bars) after finetuning on VQA.

2024a), CV-Bench 3D (Tong et al., 2024a), and RealWorldQA (xAI, 2024).

## F.6. MoE Models

All models are trained for approximately 57B tokens using a global batch size of 512 sequences (length 4096) on 64 NVIDIA H200 GPUs. The base Transformer has 16 layers with  $d_{\text{model}} = 2048$ , GQA with 32 query heads and 8 key-value heads, and FFN expansion multiplier of 1.5. For MoE layers, we use expert dimension  $d_{\text{expert}} = 512$  with top- $k = 16$  routing. Table 6 shows the parameter counts for the sparsity sweep in Sec. 6.1.

Table 6. Hyperparameters and parameter counts for the sparsity sweep.

| Configuration  | Total Params | Active Params |
|----------------|--------------|---------------|
| Dense Baseline | 1.50B        | 1.50B         |
| $E = 32$       | 2.30B        | 1.50B         |
| $E = 64$       | 3.92B        | 1.50B         |
| $E = 128$      | 7.14B        | 1.50B         |
| $E = 256$      | 13.59B       | 1.51B         |
| $E = 512$      | 26.48B       | 1.52B         |
| $E = 1008$     | 51.46B       | 1.53B         |

## F.7. Analysis on MoE Models

We perform analysis on the  $G = 16$  model (256 experts, 1 modality-specific expert) trained with SigLIP 2 and  $x$ -prediction on 1T tokens (50% text, 25% video, 22.5% image-text, 2.5% action).

**Specialization Score.** For expert  $e_i$  and modality  $m$ , we compute the selection rate  $R_m^{(i)} = C_m^{(i)} / (N_m \times k)$ . The specialization score is:  $S_i = (R_{\text{text}}^{(i)} - R_{\text{image}}^{(i)}) / (R_{\text{text}}^{(i)} + R_{\text{image}}^{(i)})$ . Experts are classified as Text ( $S_i > 0.5$ ), Vision ( $S_i < -0.5$ ), or Multimodal ( $-0.5 \leq S_i \leq 0.5$ ).

## G. Additional MoE Figures

This section presents supplementary figures for the MoE analysis in Sec. 6.1. Figure 20 details granularity and prediction-target interactions, Figure 21 shows sparsity scaling, Figures 22, 23 and 25 analyze expert specialization patterns, and Figure 24 compares training loss across encoder types.

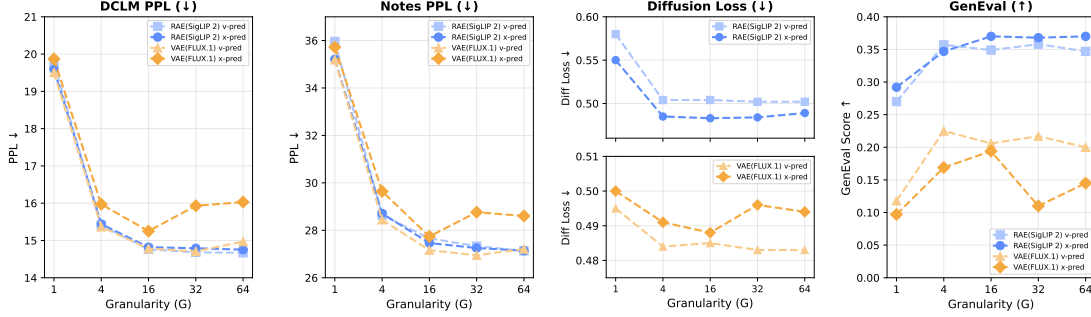


Figure 20. **Fine-grained experts improve performance;  $x$ -pred benefits high-dimensional diffusion.** Increasing granularity improves all metrics up to  $G=16$ . For RAE (SigLIP 2),  $x$ -pred (solid) outperforms  $v$ -pred (dashed) on image generation; for VAE (FLUX.1),  $v$ -pred is better.

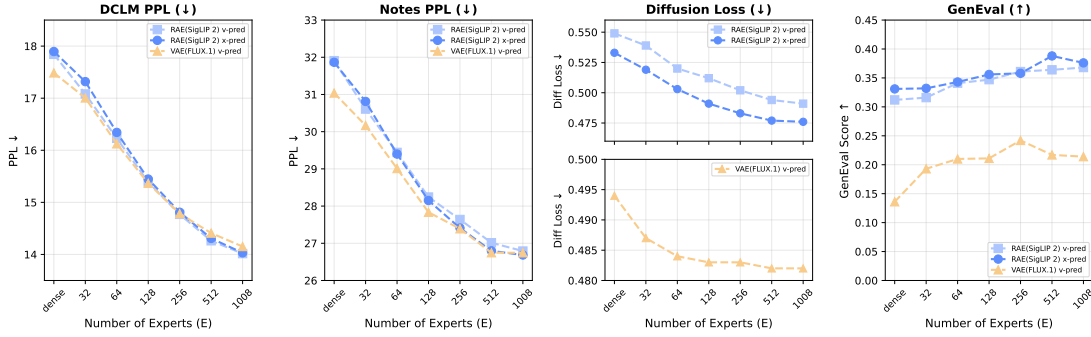


Figure 21. **Sparsity scales multimodal performance.** At fixed *active* compute (16 experts), increasing *total* expert count from 32 to 1008 consistently improves both language (lower PPL) and vision (lower diffusion loss, higher GenEval).

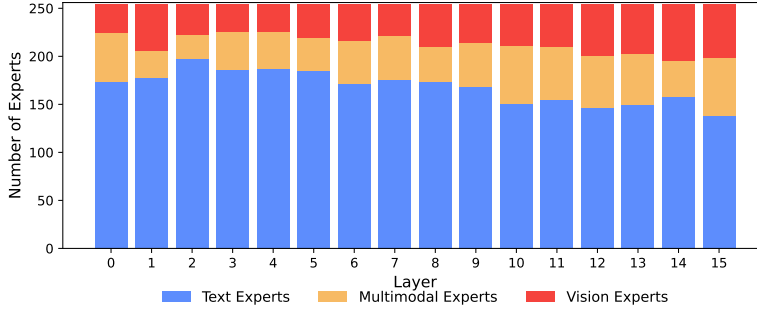


Figure 22. **Emergent Modality Specialization.** Expert specialization forms naturally. Most experts are text-focused, but later layers contain more vision and multimodal experts.

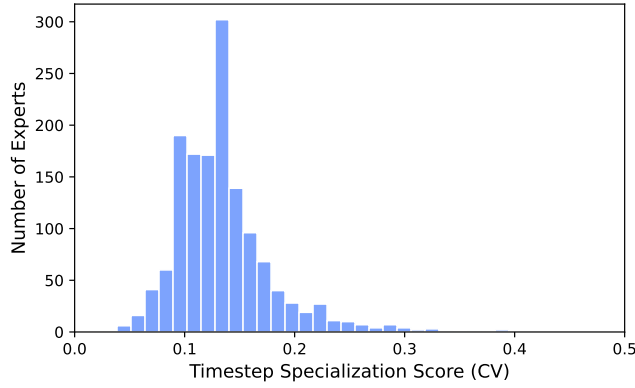


Figure 23. **Experts do not specialize in diffusion timesteps.** CV scores are distributed near 0.15, indicating vision experts are time-invariant.

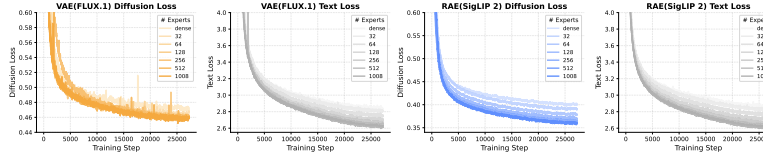


Figure 24. **Semantic representations scale better with sparsity.** We fix 16 active experts and increase the total expert count. Text loss improves consistently regardless of the vision encoder, but diffusion loss saturates for VAE (FLUX.1) while continuing to improve for RAE (SigLIP 2).

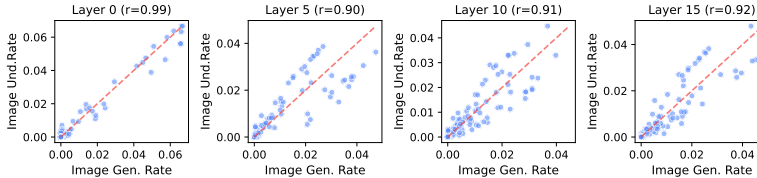


Figure 25. **Visual generation and understanding share the same experts.** We compare the expert selection rates for image generation (x-axis) versus image understanding (y-axis) across different network depths. We observe very high Pearson correlation ( $\geq r = 0.90$ ) across all layers.

## H. Additional Architecture Figures

This section contains supplementary figures for the architecture stacking analysis in Sec. 6.3. Figure 26 shows the progressive effect of each design choice, and Figure 27 evaluates world knowledge retention across architectures.

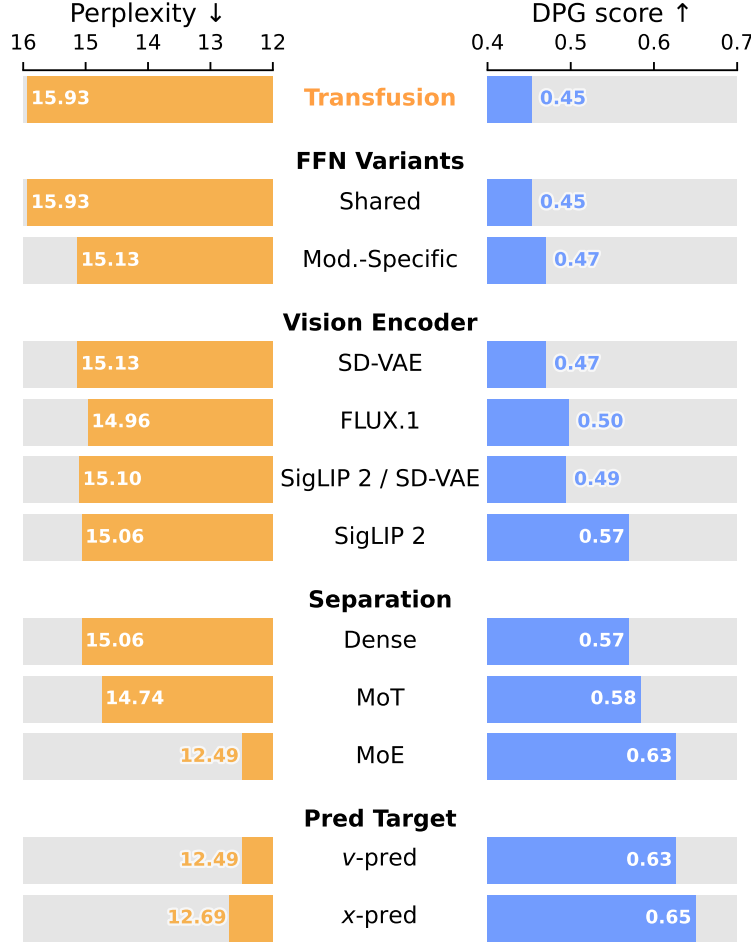


Figure 26. **Modality-specific FFN, SigLIP 2, MoE, and  $x$ -pred achieve best unified performance.** Starting from a Transfusion baseline, we progressively stack the best design choices for FFN, vision encoder, capacity separation, and prediction target. Each row shows **Perplexity** (↓) and **DPG score** (↑).

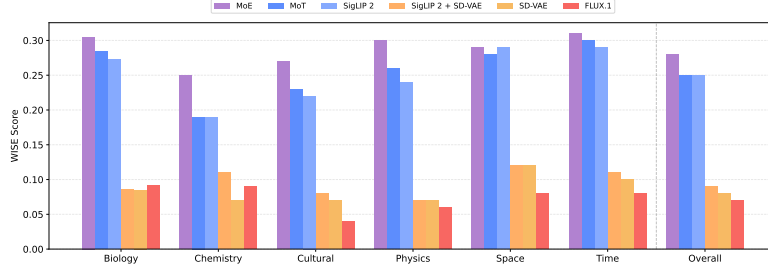


Figure 27. **Semantic encoders and MoE capture more world knowledge.** WISE scores across six knowledge categories and overall. Models with semantic encoders (MoE, MoT, SigLIP 2) consistently outperform VAE-based models (SD-VAE, FLUX.1) by 3–4 $\times$ , with MoE achieving the highest overall score.

## I. World Modeling

This section provides implementation details and additional qualitative results for the world modeling experiments in Sec. 5. Figure 28 illustrates the data format.

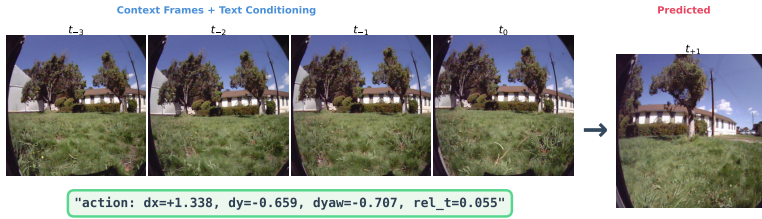


Figure 28. **Navigation actions are encoded directly as text tokens.** A training sequence consists of four context frames and a navigation action formatted as natural language text. The model predicts the resulting target view.

### I.1. Implementation Details

**CEM planning.** We implement the Cross-Entropy Method following NWM (Bar et al., 2025). The planner uses an 8-step horizon, samples  $N=120$  candidate action sequences, and predicts 8 future frames per sequence. Trajectories are scored via LPIPS distance between the final predicted frame and the goal image. We select the top  $K=5$  trajectories and average their actions.

**Data ratio studies.** We vary the proportion of NWM data in the training mixture. The base mixture allocates 25% text, 25% video, and 50% multimodal. We ablate NWM data from 0.1% to 25% of total budget.

**WASD control definition.** We map keyboard controls to robot-centric coordinates (Figure 29): **Forward** (W):  $(+0.5, 0, 0)$ , **Backward** (S):  $(-0.5, 0, 0)$ , **Turn Left** (A):  $(+0.2, 0, +0.5)$ , **Turn Right** (D):  $(+0.2, 0, -0.5)$ .

### I.2. More Navigation Examples

We showcase additional navigation rollouts with either WASD controls or free-form natural language as actions in Figures 30 to 34. Figure 35 demonstrates counterfactual trajectories generated from the same context with different action sequences.

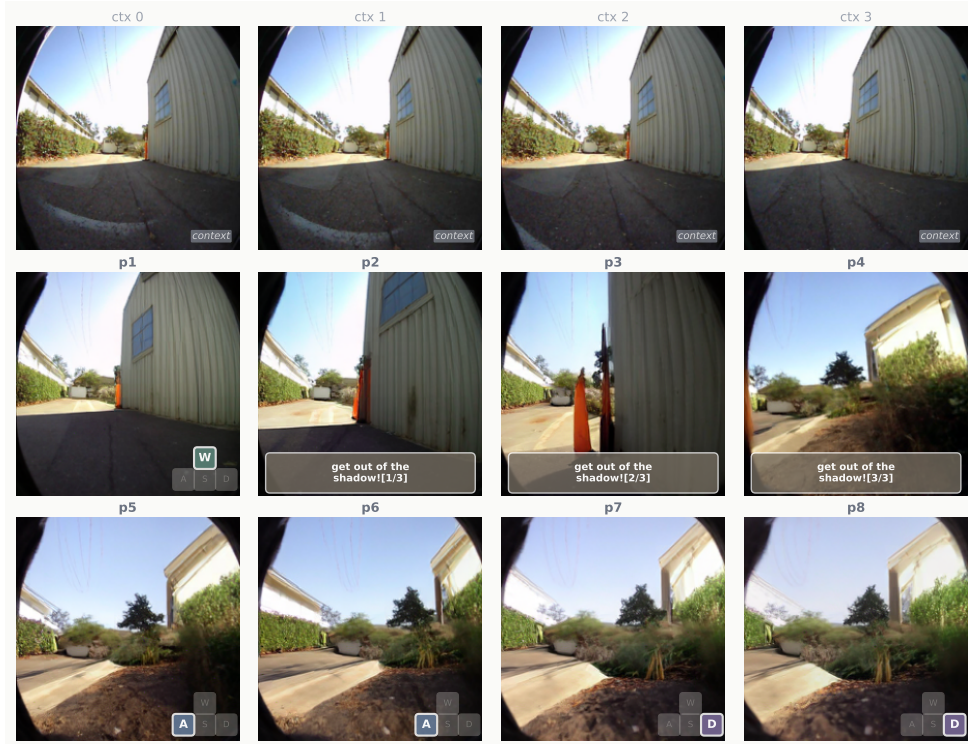


Figure 29. **Zero-shot model controllability via natural language.** Given four context frames (ctx0–ctx3), the model predicts eight future frames (p1–p8) conditioned on WASD-style or natural language navigation actions.



Figure 30. **More navigation examples with natural language commands (I).**

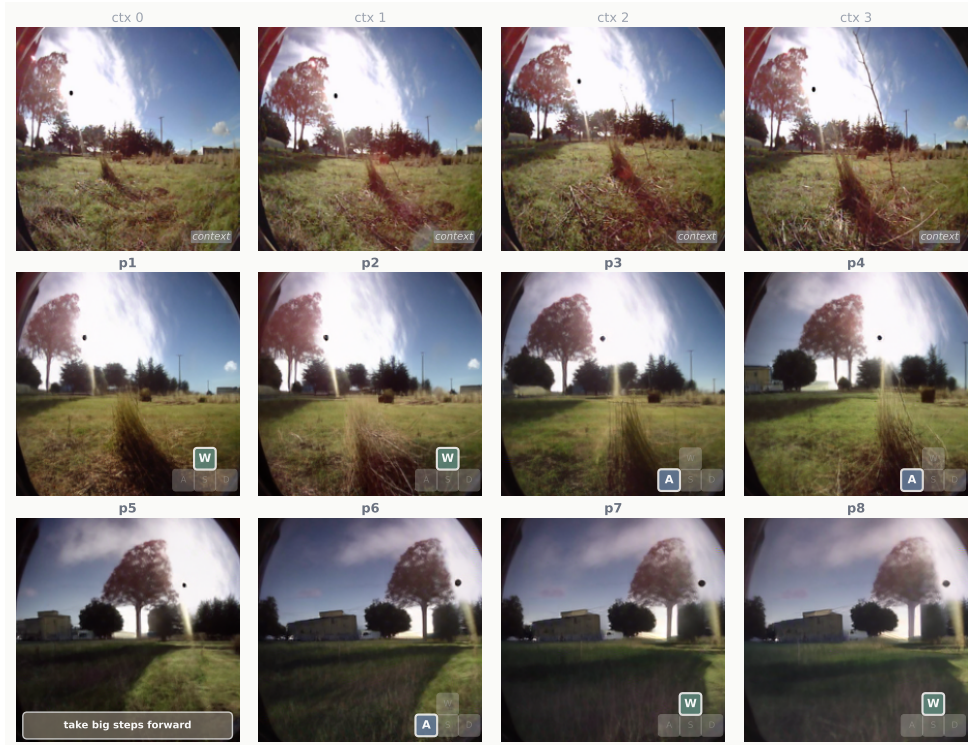


Figure 31. More navigation examples with natural language commands (II).

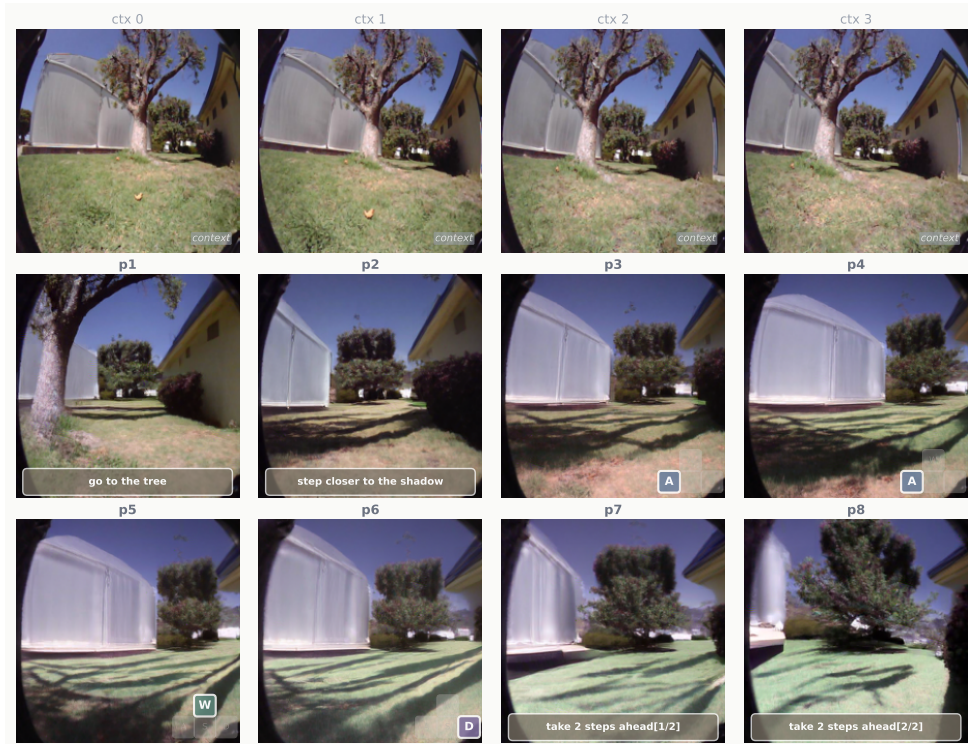


Figure 32. More navigation examples with natural language commands (III).

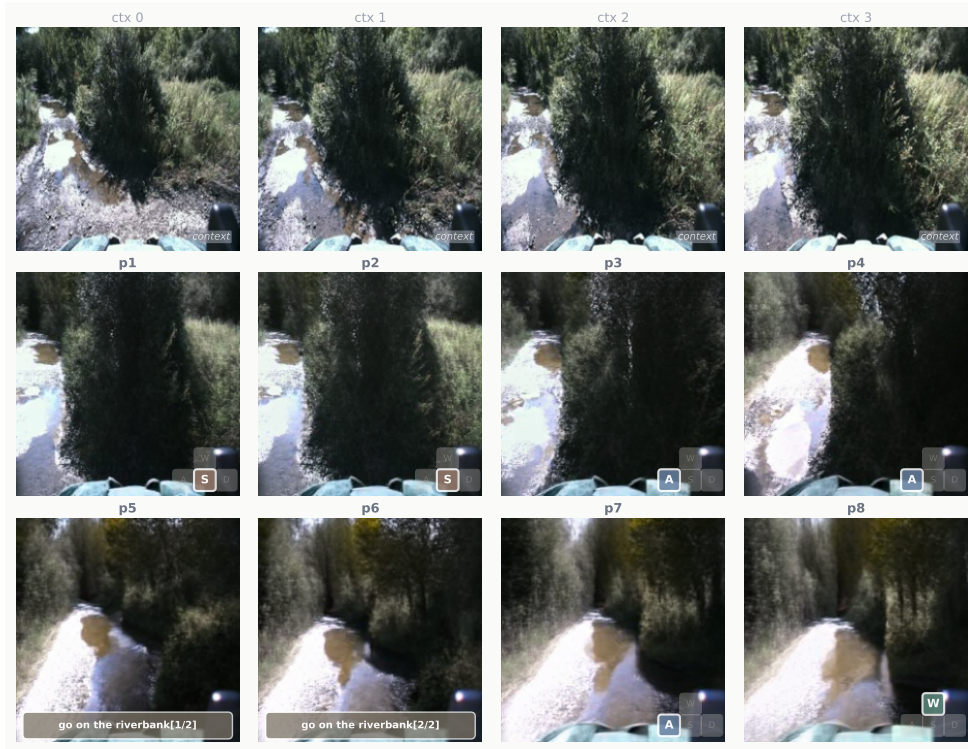


Figure 33. More navigation examples with natural language commands (IV).



Figure 34. More navigation examples with natural language commands (V).



Figure 35. Counterfactual navigation trajectories from the same context. The model generates counterfactual trajectories following different action sequences.